

	M/M/1	M/M/s>1	M/M/1/K
Probabilidade do sistema estar ocioso	$P_0 = 1 - \rho$	$P_0 = 1 / \left[\sum_{n=0}^{s-1} \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \times \frac{1}{1 - \lambda/(s\mu)} \right] \quad \rho = \lambda/s\mu$	$P_0 = \frac{1 - \rho}{1 - \rho^{K+1}} \quad \rho = \lambda/\mu$
Probabilidade de haver n clientes no sistema	$P(n) = \left(\frac{\lambda}{\mu}\right)^n \cdot \left(\frac{\mu - \lambda}{\mu}\right) = (1 - \rho)\rho^n$	$\begin{array}{ll} n \leq S & n \geq S \\ P_n = \frac{(\lambda/\mu)^n}{n!} \times P_0 & P_n = \frac{(\lambda/\mu)^n}{s! s^{n-s}} \times P_0 \end{array}$	$P_n = \frac{1 - \rho}{1 - \rho^{K+1}} \rho^n$
Número médio de clientes no sistema (L)	$L = \frac{\lambda}{\mu - \lambda}$	$L = L_q + \frac{\lambda}{\mu}$	$L = \frac{\rho}{1 - \rho} - \frac{(K+1)\rho^{K+1}}{1 - \rho^{K+1}}$
Número médio de clientes na fila (Lq)	$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$	$L_q = \frac{P_0(\lambda/\mu)^s \rho}{s!(1 - \rho)^2}$	$L_q = L - (1 - P_0)$
Tempo médio gasto no sistema (W)	$W = \frac{1}{(\mu - \lambda)}$	$W = \frac{L}{\lambda} = W_q + \frac{1}{\mu}$	$W = \frac{L}{\bar{\lambda}} \rightarrow \bar{\lambda} = \sum_{n=0}^{K-1} \lambda P_n = \lambda(1 - P_K)$
Tempo médio gasto na fila (Wq)	$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$	$W_q = \frac{L_q}{\lambda}$	$W_q = \frac{L_q}{\bar{\lambda}} \rightarrow \bar{\lambda} = \sum_{n=0}^{K-1} \lambda P_n = \lambda(1 - P_K)$
Probabilidade (n>r)	$P(n > r) = \left(\frac{\lambda}{\mu}\right)^{r+1}$		
Probabilidade (W>t)	$P(W > t) = e^{-\mu(1-\rho)t}$	$P(W > t) = e^{-\mu t} \left[1 + \frac{P_0(\lambda/\mu)^s}{s!(1 - \rho)} \left(\frac{1 - e^{-\mu t(s-1-\lambda/\mu)}}{s-1-\lambda/\mu} \right) \right]$	
Probabilidade (Wq>t)	$P(W_q > t) = \rho e^{-\mu(1-\rho)t}$	$P(W_q > t) = [1 - P(W_q = 0)] e^{-s\mu(1-\rho)t}$, em que $P(W_q = 0) = \sum_{n=0}^{s-1} P_n$	

	M/M/s>1/K	M/M/1/N	M/M/s>1/N
P(n=0)	$P_0 = 1 / \left[\sum_{n=0}^s \frac{(\lambda/\mu)^n}{n!} + \frac{(\lambda/\mu)^s}{s!} \times \sum_{n=s+1}^K \left(\frac{\lambda}{s\mu} \right)^{n-s} \right] \quad \rho = \left(\frac{\lambda}{s\mu} \right)$	$P_0 = 1 / \left[\sum_{n=0}^N \frac{N!}{(N-n)!} \left(\frac{\lambda}{\mu} \right)^n \right] \quad \rho = \left(\frac{N\lambda}{\mu} \right)$	$P_0 = 1 / \left[\sum_{n=0}^{s-1} \frac{N!}{(N-n)! n!} \left(\frac{\lambda}{\mu} \right)^n + \sum_{n=s}^N \frac{N!}{(N-n)! s! s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n \right] \quad \rho = \left(\frac{N\lambda}{s\mu} \right)$
P(n clientes no sistema)	$\begin{array}{cc} n \leq S & s \leq n \leq K \\ P_n = \frac{(\lambda/\mu)^n}{n!} \times P_0 & P_n = \frac{(\lambda/\mu)^n}{s! s^{n-s}} \times P_0 \end{array}$	$P_n = \frac{N!}{(N-n)!} \left(\frac{\lambda}{\mu} \right)^n \times P_0 \quad se \ n = 1, 2, \dots, N$	$\begin{array}{cc} n \leq S & s \leq n \leq N \\ P_n = \frac{N!}{(N-n)! n!} \left(\frac{\lambda}{\mu} \right)^n \times P_0 & P_n = \frac{N!}{(N-n)! s! s^{n-s}} \left(\frac{\lambda}{\mu} \right)^n \times P_0 \end{array}$
L	$L = \sum_{n=0}^{s-1} n P_n + L_q + s \left(1 - \sum_{n=0}^{s-1} P_n \right)$	$L = N - \frac{\mu}{\lambda} (1 - P_0)$	$L = \sum_{n=1}^N n P_n$
Lq	$L_q = \frac{P_0 (\lambda/\mu)^s \rho}{s! (1 - \rho)^2} [1 - \rho^{K-s} - (K-s) \rho^{K-s} (1 - \rho)]$	$L_q = N - \left(\frac{\lambda + \mu}{\lambda} \right) (1 - P_0)$	$L_q = L - \left(\frac{\lambda}{\mu} \right) (N - L)$
W	$W = \frac{L}{\bar{\lambda}} \rightarrow \bar{\lambda} = \sum_{n=0}^{K-1} \lambda P_n = \lambda (1 - P_K)$	$W = \frac{L}{\bar{\lambda}} \rightarrow \bar{\lambda} = \lambda (N - L)$	$W = \frac{L}{\bar{\lambda}} \rightarrow \bar{\lambda} = \lambda (N - L)$
Wq	$W_q = \frac{L_q}{\bar{\lambda}} \rightarrow \bar{\lambda} = \sum_{n=0}^{K-1} \lambda P_n = \lambda (1 - P_K)$	$W_q = \frac{L_q}{\bar{\lambda}} \rightarrow \bar{\lambda} = \lambda (N - L)$	$W_q = \frac{L_q}{\bar{\lambda}} \rightarrow \bar{\lambda} = \lambda (N - L)$

	M/M/s					M/M/2/k					M/M/1/N				M/M/2/N			
N/μ	s=2	s=3	k=2	k=3	k=4	N/μ	N=2	N=3	N=4	N=2	N=3	N=4	N=2	N=3	N=4	N=2	N=3	N=4
0,2	0,8182	0,8187	0,8197	0,8183	0,8182	0,2	0,6757	0,5297	0,3983	0,6944	0,5774	0,4778	0,6944	0,5774	0,4778	0,6944	0,5774	0,4778
0,4	0,6667	0,6701	0,6757	0,6684	0,6670	0,4	0,4717	0,2822	0,1499	0,5102	0,3602	0,2487	0,5102	0,3602	0,2487	0,5102	0,3602	0,2487
0,6	0,5385	0,5479	0,5618	0,5453	0,5405	0,6	0,3425	0,1598	0,0624	0,3906	0,2379	0,1380	0,3906	0,2379	0,1380	0,3906	0,2379	0,1380
0,8	0,4286	0,4472	0,4717	0,4448	0,4349	0,8	0,2577	0,0970	0,0294	0,3086	0,1643	0,0810	0,3086	0,1643	0,0810	0,3086	0,1643	0,0810
1,0	0,3333	0,3636	0,4000	0,3636	0,3478	1,0	0,2000	0,625	0,0154	0,2500	0,1176	0,0500	0,2500	0,1176	0,0500	0,2500	0,1176	0,0500